Midterm prep guide (Group 9) (Otar Jintchveladze)

- 45 minutes
- Algorithmic Patterns preferred.
- Manually implement the logic using loops (must be completely correct to earn points).

*“I would suggest reading the task at least twice, identifying attributes and methods carefully!*

*Do not get nervous, relax and do the task like you did during the practices. There will be no surprises and tricks on the exam. You will be examined only in the things that we have done in the class!*

*In case of questions about the task do not hesitate to raise your hand and ask, that will prevent any misunderstandings regarding the task!”*

**Core topics to pass:**
- Classes & Objects.
- Attributes & Methods.
- Encapsulation.
- Relationships (Composition, Aggregation).
- Algorithmic Patterns

*“If you do not understand any of these topics, I suggest going back to the lecture notes and revising them”*

**How to practice:**
- Solve Ranger Task and Cake Shop task from week 3. (Similar to midterm).
- Solve Library Task from week 5. (Similar to midterm, implements every algo pattern)
- If you still feel not confident, go over assignments and try doing one or two from there. (That should be already more than enough)

*"The biggest challenge for most students is implementing algorithmic patterns. As a teacher, I can explain the concepts and provide guidance, but at the end of the day mastering algorithmic thinking depends on how much a student practices. It's a skill that only becomes **intuitive** through consistent effort."*

**Algorithmic Patterns :**

- Summation
  - e.x SUMMATION(t, f(e) -> e, cond(e) -> true, op(a, b) -> a + b, identity -> 0)
- Counting
  - e.x COUNTING(t, cond(e) -> e % 2 = 0)
- Linear Search
  - e.x SEARCH(t, cond(e) -> e = 9)
- Maximum Selection
  - e.x MAX(t, f(e) -> e)
- Conditional Maximum Search
  - e.x CONDMAX(t, f(e) -> e, cond(e) -> e % 2 = 0)